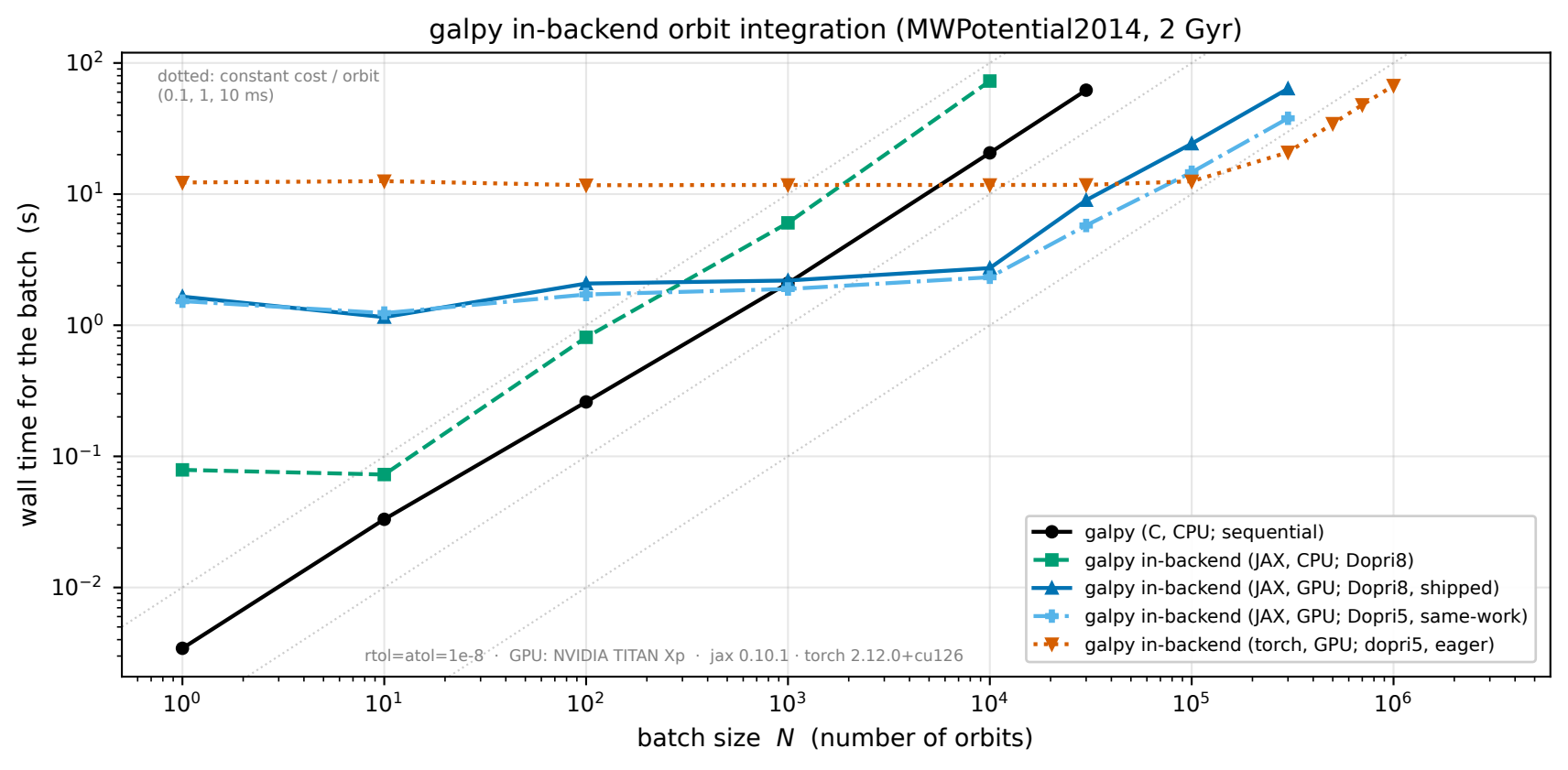
galpy in-backend orbit integration (MWPotential2014, 2 Gyr)
dotted: constant cost / orbit
(0.1, 1, 10 ms)
wall time for the batch (s)
batch size N (number of orbits)
rtol=atol=1e-8 · GPU: NVIDIA TITAN Xp · jax 0.10.1 · torch 2.12.0+cu126
galpy (C, CPU; sequential)
galpy in-backend (JAX, CPU; Dopri8)
galpy in-backend (JAX, GPU; Dopri8, shipped)
galpy in-backend (JAX, GPU; Dopri5, same-work)
galpy in-backend (torch, GPU; dopri5, eager)